cially the rotator cuff mechanism and subscapularis), the type and quality of the surgical procedure(s) performed, and patient-related factors, such as participation in a postoperative rehabilitation program.[32,36,195] The outcomes most often measured in follow-up studies using a variety of quantitative scales are pain relief and other paramenters of quality of life, passive and active shoulder ROM, and the ability to perform functional activities.

Despite the emphasis in numerous resources that a patient's participation in postoperative rehabilitation is crucial for successful outcomes, there are no studies to support this opinion because all patients undergoing shoulder arthroplasty are given some form of postoperative exercise instruction. Furthermore, published protocols are routinely modified to meet the needs of individual patients and consequently have not been compared to determine if one protocol yields better outomes than another.[195]

Pain relief. A decrease in pain is the most dramatic result of glenohumeral arthroplasty. Almost all patients—regardless of the underlying pathology, the type of arthroplasty (TSR or hemireplacement), or the design of the prosthetic implants—report complete or substantial relief of shoulder pain and improved functional use of the arm.[36,108,109,117,119,121,130,134,171]

The extent of pain relief has been shown to be associated with the underlying cause(s) of glenohumeral arthritis. Neer et al.,[119] Matsen,[108] and more recently Norris and Iannotti[130] reported that 90% of patients with primary OA or osteonecrosis had complete or near-complete pain relief after TSR. Similar results have been reported for patients with OA who underwent hemiarthroplasty.[101,109,121] Patients with RA or other synovium-based diseases also report substantial pain relief after TSR or hemiarthroplasty, although not quite to the extent reported by patients with OA or osteonecrosis.[36,159,182]

Whether TSR is more effective than hemiarthroplasty for pain relief has also been studied. In a prospective follow-up study over a mean of 4.3 years when patients with OA having TSR were compared to those having hemiarthroplasty, postoperative pain scores were reported to be similar in the two groups, with patients in the TSR group demonstrating more improvement because of a higher level of pain preoperatively.[134] In another study, patients with OA were randomly assigned to undergo either TSR or hemiarthroplasty and were evaluated postoperatively over a 24-month period. Results of this study indicated that both groups of patients reported significant pain relief and improvements in other quality-of-life parameters, with no significant differences between the TSR and hemiarthroplasty groups.[101] Whether TSR versus hemiarthroplasty is more effective for pain relief in patients with RA has not been clearly established.[109,182]

ROM and functional use of the upper extremity. Despite the emphasis placed on improving ROM and use of the arm for function activities during rehabilitation after shoulder arthroplasty, improvements in these outcomes are less predictable than pain relief, with the functional status improving more consistently than ROM.* In general, patients with primary OA or osteonecrosis demonstrate greater improvement in active ROM (forward elevation and shoulder rotation) than patients with RA, in part because of a higher incidence of cuff deficiency asociated with RA or the use of more constrained prosthetic designs.[159,182,195] For example, in patients with OA or osteonecrosis, the mean active forward elevation of the shoulder (reported in reviews of a number of studies) changed from 105° to 161°. In patients with RA means ranged from 75° to 105°.[171,195]

Significant improvement in functional status has been reported for patients with OA or osteonecrosis. Although functional improvement after arthroplasty has been reported for patients with RA, many studies used nonstandardized measurement tools, making it difficult to compare their results with those of other studies.[195] Regardless of the underlying pathology, resources agree that a well functioning rotator cuff mechanism is the basis for significant postoperative gains in active ROM and functional abilities.[36,168,195]

Arthrodesis of the Shoulder

Indications for Surgery

The following are generally accepted indications for arthrodesis of the GH joint.[109,145,146,196]

- Incapacitating pain (at rest or with movement)
- Gross instability of the GH joint
- Complete paralysis of the deltoid and rotator cuff muscles
- Severe joint destruction due to infection
- Failed shoulder arthroplasty in a young, active patient who is not a candidate for revision arthroplasty
- Sufficient compensatory scapular motion and strength of the serratus anterior and trapezius muscles

Procedure

The GH joint is fused with pins and bone grafts in a position of 15° to 30° flexion and abduction and up to 30° to 45° internal rotation so the hand can reach the middle of the body or the mouth.[109,145,146,174,196] The shoulder is immobilized in a shoulder spica cast or a thoracobrachial (airplane) splint that extends across the elbow joint. The immobilizer is worn for 3 to 5 months.

Postoperative Management

Place emphasis on maintaining mobility of peripheral joints (wrist and hand) while the shoulder and elbow are immobilized. If a splint with a hinged elbow joint is used, begin elbow flexion and extension when permis-

*See references 57, 109, 130, 134, 161, 171, 195.

sible, often as early as the day after surgery. After the immobilization device may be removed for exercise, begin active scapulothoracic ROM. Strengthen the scapulothoracic musculature to maximize control and stability of the scapula.

Outcomes

A patient may expect to achieve 90° to 130° of active elevation of the arm because of scapulothoracic mobility.[109,145,146,196] After bony and soft tissue healing is complete, the shoulder is stable and pain-free for activities that require strength or weight bearing at the shoulder. Patients are able to bring the hand to the mouth, behind the head, and to the hip.[145,174,196] Over time, excessive stress may be placed on the AC joint, resulting in AC joint hypermobility and pain.[145,146,174]

Painful Shoulder Syndromes (Rotator Cuff Disease, Impingement Syndromes, Shoulder Instabilities): Nonoperative Management

Mechanical compression and irritation of the soft tissues (rotator cuff and subacromial bursa) in the suprahumeral space (see Fig. 17.7) is called *impingement syndrome* and is the most common cause of shoulder pain.[99,105] Various etiologic factors have been identified and therefore have led to several classification systems, which are summarized in Box 17.7.

Related Pathologies and Etiology of Symptoms

The cause of impingement is multifactoral, involving both structural and mechanical impairments. Impingement syndrome is often used as the diagnosis when the patient's signs and symptoms are related to pain with overhead reaching, a painful arc mid-range, and positive impingement tests. Other test results may more specifically identify the tissue involved, the faulty mechanics associated with the condition, or the degree of instability or injury. Symptoms that derive from impingement are usually brought on with excessive or repetitive overhead activities that load the shoulder joint, particularly in the mid-range. Impingement syndromes are generally classified as *primary* or *secondary*.

Other types of musculotendinous strain in the shoulder region may result from overuse or trauma, such as in the anterior pectoral region from racket sports or in the long head of the triceps and serratus anterior from impact trauma such as holding onto a steering wheel in an automobile accident.

BOX 17.7 Categories of Painful Shoulder Syndromes

Impingement syndromes and other painful shoulder conditions have varying etiologic factors and therefore can be categorized several ways.

Based on Degree or Stage of Pathology of the Rotator Cuff (Neer's Classification of Rotator Cuff Disease)[118]

- *Stage I.* Edema, hemorrhage (patient usually < 25 years of age)
- *Stage II.* Tendinitis/bursitis and fibrosis (patient usually 25 to 40 years of age)
- *Stage III.* Bone spurs and tendon rupture (patient usually > 40 years of age)

Based on Impaired Tissue[41]

- Supraspinatus tendinitis
- Infraspinatus tendinitis
- Bicipital tendinitis
- Subdeltoid (subacromial) bursitis
- Other musculotendinous strains (specific to type of injury or trauma)
  - Anterior—from overuse with racket sports (pectoralis minor, subscapularis, coracobrachialis, short head of biceps strain)
  - Inferior—from motor vehicle trauma (long head of triceps, serratus anterior strain)

Based on Mechanical Disruption and Direction of Instability or Subluxation

- Multidirectional instability from lax capsule with or without impingement
- Unidirectional instability (anterior, posterior, or inferior) with or without impingement
  - Traumatic injury with tears of capsule and/or labrum
  - Insidious (atraumatic) onset from repetitive microtrauma
  - Inherent laxity

Based on Progressive Microtrauma (Jobe's classification)[88]

- *Group 1.* Pure impingement (usually in an older recreational athlete with partial undersurface rotator cuff tear and subacromial bursitis)
- *Group 2.* Impingement associated with labral and/or capsular injury, instability, and secondary impingement
- *Group 3.* Hyperelastic soft tissues resulting in anterior or multidirectional instability and impingement (usually attenuated but intact labrum, undersurface rotator cuff tear)
- *Group 4.* Anterior instability without associated impingement (result of trauma; results in partial or complete dislocation)

Based on Degree and Frequency

- Instability → subluxation → dislocation
- Acute, recurrent, fixed

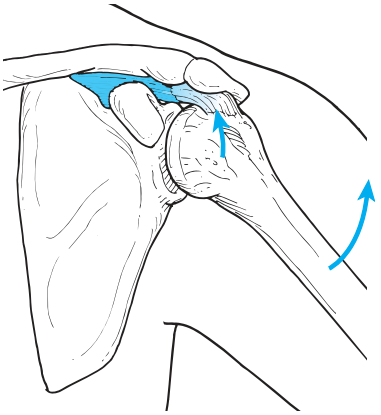


FIGURE 17.15 Decrease in the suprahumeral space during repetitive elevation activities lead to symptoms of impingement.

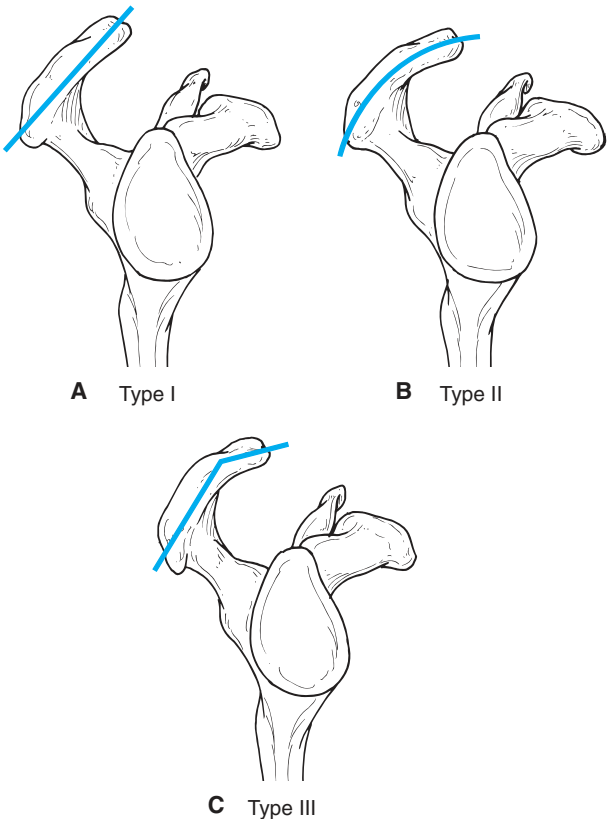


FIGURE 17.16 Classifications of the acromion by shape: (*A*) type I (flat); (*B*) type II (curved); (*C*) type III (hooked).

Primary Impingement—Rotator Cuff Disease

Primary impingement is believed to occur as a result of mechanical wear of the rotator cuff against the anteroinferior one-third of the acromion in the suprahumeral space during elevation activities of the humerus (Fig. 17.15). Encroachment may be the result of *intrinsic* or *extrinsic* factors.

Intrinsic factors. Intrinsic factors are those that are directly associated with encroachment of the subacromial space. They include vascular changes in the rotator cuff tendons, structural variations in the acromion, hypertrophic degenerative changes of the AC joint, or other trophic changes in the coracoacromial arch or humeral head. All of these factors decrease the suprahumeral space and often have to be dealt with surgically.[58,82,149]

Neer[120] first suggested that in patients undergoing surgery for impingement syndrome the size and shape of the structures that make up the coricoacromial arch are related to rotator cuff impingement. In later studies variations in the shape and thickness of the acromion were identified and classified. One such classification, based on the shape of the anteroinferior aspect of the acromion, identified three shapes: type I (flat), type II (curved), and type III (hooked) (Fig. 17.16). Rotator cuff pathology is often associated with type II and III, but not type I, acromial shapes.[1,115]

Extrinsic factors. Factors that result in decreased suprahumeral space and repetitive trauma to the soft tissues during elevation of the arm include posterior capsular tightness, poor neuromuscular control of the rotator cuff or scapular muscles, faulty scapulothoracic posture with muscle imbalances, or a partial or complete tear of the tissues in the suprahumeral space (incurred during a traumatic or degenerative situation).[74,99,103,105,118,167]

Neer described three pathological stages of impingement that demonstrate the potentially progressive nature of the pathology over time (see Box 17.7).[118] Other authors have identified chronic inflammation, possibly due to repetitive microtrauma in the joint region, as a stimulus for the development of frozen shoulder.[66,122,125]

Tendinitis/Bursitis

Neer identified tendinitis/bursitis as a stage II impingement syndrome (see Box 17.7). The following sections describe specific pathological diagnoses and presenting signs and symptoms.

Supraspinatus tendinitis. With supraspinatus tendinitis, the lesion is usually near the musculotendinous junction, resulting in a painful arc with overhead reaching. Pain occurs with the impingement test (forced humeral elevation in the plane of the scapula while the scapula is passively stabilized so the greater tuberosity impacts against the acromion[74,118] or with the arm in internal rotation while flexing the humerus).[74] There is pain on palpation of the tendon just inferior to the anterior aspect of the acromion when the patient's hand is placed behind the back. It is difficult to differentiate tendinitis from subdeltoid bursitis because of the anatomical proximity.

Infraspinatus tendinitis. With infraspinatus tendinitis, the lesion is usually near the musculotendinous junction,

resulting in a painful arc with overhead or forward motions. It may present as a deceleration (eccentric) injury due to overload during repetitive or forceful throwing activities. Pain occurs with palpation of the tendon just inferior to the posterior corner of the acromion when the patient horizontally adducts and laterally rotates the humerus.

Bicipital tendinitis. With bicipital tendinitis, the lesion involves the long tendon in the bicipital groove beneath or just distal to the transverse humeral ligament. Swelling in the bony groove is restrictive and compounds and perpetuates the problem. Pain occurs with resistance to the forearm in a supinated position while the shoulder is flexing (Speed's sign) and on palpation of the bicipital groove.[106] Rupture or dislocation of this humeral depressor may escalate impingement of tissues in the suprahumeral space.[118,126]

Bursitis (subdeltoid or subacromial). When acute, the symptoms of bursitis are the same as those seen with supraspinatus tendinitis. Once the inflammation is under control, there are no symptoms with resistance.

Other Impaired Musculotendinous Tissues

The following are examples of other musculotendinous problems in the shoulder region.

- The pectoralis minor, short head of the biceps, and coracobrachialis are subject to microtrauma, particularly in racket sports requiring a controlled backward then a rapid forward swinging of the arm, as are the scapular stabilizers as they function to control forward motion of the scapula.[100]
- The long head of the triceps and scapular stabilizers are often injured in motor vehicle accidents as the driver holds firmly to the steering wheel on impact.
- A fall on an outstretched hand or against the shoulder may also cause trauma to the scapular stabilizers, which, if not properly healed, continue to cause symptoms whenever using the arm or when maintaining a shoulder posture.
- Injury, overuse, or repetitive trauma can occur in any muscle being subjected to stress.[129] Pain occurs when the involved muscle is placed on a stretch or when contracting against resistance. Palpating the site of the lesion causes the familiar pain.

Secondary impingement—Shoulder Instability/Subluxation

Secondary impingement is used to describe symptoms resulting from faulty mechanics due to hypermobility or instability of the GH joint with increased translation of the humeral head. The instability may be *multidirectional* or *unidirectional*.

Multidirectional instability. Some individuals have physiologically lax connective tissue causing excessive mobility in the joints of the body. In the GH joint, the humeral head translates to a greater degree than normal in all directions.[133,160,178] Many individuals, particularly those involved in overhead throwing or lifting activities, have some inherent laxity or develop laxity of the capsule and instability from continually subjecting the joint to stretch forces.[58,88] A hypermobile joint may be satisfactorily supported by strong rotator cuff muscles; but once the muscles fatigue, poor humeral head stabilization leads to faulty humeral mechanics, trauma, and inflammation of the suprahumeral tissues.[88,114] This trauma is magnified with the rapidity of control demanded in the overhead throwing action.[58] Similarly, in individuals with poor rotator cuff muscle strength and function, the ligaments become stressed with repetitive use and hypermobility, resulting in impingement. With instability, the impingement of tissue in the suprahumeral space is the secondary effect.[58]

Hypermobility can cause problems in addition to impingement, such as subluxation, dislocation, or rotator cuff tendinitis, which with repetitive microtrauma can lead to degenerative changes including bone spurs, tendon rupture, or capsular restrictions and frozen shoulder. Jobe developed a classification system incorporating the progression of impingement and instability in the overhead athlete based on progressive microtrauma[88] (outlined in Box 17.7).

Unidirectional instability with or without impingement. Unidirectional instability (anterior, posterior, or inferior) may be the result of physiologically lax connective tissue but is usually the result of trauma and usually involves rotator cuff tears. The tears can be classified as acute, chronic, degenerative, or partial- or full-thickness tears. Often there is damage to the glenoid labrum and tearing of some of the supporting ligaments.

Traumatic tears or paralysis. Partial-thickness tears or full avulsion of the greater tubercle may occur in the elderly as the result of falling on an outstretched arm.[123] In young patients, trauma is usually associated with capsular injury, with or without labrum injury, resulting in instability. Dislocation of the humerus may occur with ensuing instability. The instability can lead to progressive degeneration and eventually tears in the supporting structures. Tears are associated with pain and most commonly weakness of shoulder abduction and external rotation.

- *Anterior instability* usually occurs with force against the arm when it is in an abducted and externally rotated position, and it frequently involves detachment of the anterior capsule and glenoid labrum (Bankart lesion). There may also be a fractured piece or flattening of the anterior lip of the glenoid.[68] Positive signs include apprehension, load and shift, and anterior drawer tests.[106,201]
- *Posterior instability* is the result of a forceful thrust against a forward-flexed humerus or fall on an outstretched arm. There is a positive posterior drawer sign.[106,201]
- *Inferior instability* is the result of rotator cuff weakness/paralysis and is frequently seen in patients with hemiplegia.[64] It is also prevalent in patients with multidirectional instability. This is detected with a positive sulcus sign.[106,201]

Insidious (atraumatic) onset. Neer has identified rotator cuff tears as a stage III impingement syndrome, a condition that typically occurs in persons over age 40 after repetitive microtrauma to the rotator cuff or long head of the biceps.[118] With aging, the distal portion of the supraspinatus tendon is particularly vulnerable to impingement or stress from overuse strain. With degenerative changes, calcification and eventual tendon rupture may occur.[58,132] Chronic ischemia caused by tension on the tendon and decreased healing in the elderly are possible explanations, although Neer stated that, in his experience, 95% of tears are initiated by impingement wear rather than by impaired circulation or trauma.[118]

Common Impairments

Various impairments have been reported to be common in impingement syndromes, although it is not known if they are the cause or effect of the faulty mechanics.[31,103,105,135,192] A thorough examination of the cervical spine and shoulder girdle is necessary to differentiate signs and symptoms related to primary and secondary impingements or other causes of shoulder pain.[22,46,106]

Impaired Posture and Muscle Imbalances

Increased thoracic kyphosis, forward head, abducted and forward-tipped scapula are often identified as related to impingement syndrome. Faulty scapular alignment may be one factor in decreasing the suprahumeral space and therefore leading to irritation of the rotator cuff tendons with overhead activities.[103] Faulty upper quadrant posture leads to an imbalance in the length and strength of the scapular and GH musculature and decreases the effectiveness of the dynamic and passive stabilizing structures of the GH joint.[202]

Typically with increased thoracic kyphosis, the scapula is protracted and tipped forward, and the GH joint is in an internally rotated posture. With this posture, the pectoralis minor, levator scapulae, and shoulder internal rotators are tight; and the lateral rotators of the shoulder and upward rotators of the scapula test weak and have poor muscular endurance. There is no longer the stabilizing tension on the superior joint capsule and coracohumeral ligament or compressive forces from the rotator cuff muscles. Consequently, the effect of gravity tends to cause an inferior force on the humerus. When reaching overhead, there is faulty scapular and humeral mechanics, resulting in faulty alignment of the scapula and altered function in the muscles controlling the scapula and GH joint.

Focus on Evidence

In a study which examined the kinematics of 52 subjects (26 without shoulder impairment and 26 with shoulder impingement), Ludewig and Cook[103] documented delayed upward rotation of the scapula during the 31° to 60° range of humeral elevation, incomplete backward tipping of the scapula, and excessive scapular elevation in individuals with impingement compared with those without shoulder impairments. This mechanical alteration may contribute to decreased clearance under the anterior acromion. They also documented decreased activation of the lower serratus anterior and overuse of the upper trapezius with scapular elevation, which was suggested as a possible compensation for the weak posterior tipping action of the serratus anterior.[103]

Decreased Thoracic ROM

Thoracic extension is a component motion that is needed for full overhead reaching. Incomplete thoracic extension decreases the functional range of humeral elevation.

Rotator Cuff Overuse and Fatigue

If the rotator cuff musculature or long head of the biceps fatigue from overuse, they no longer provide the dynamic stabilizing, compressive, and translational forces that support the joint and control the normal joint mechanics. This is thought to be a precipitating factor in secondary impingement syndromes when there is capsular laxity and increased need for muscular stability.[137] The tissues in the subacromial space may then become impinged as a result of faulty mechanics. There is also a relationship between muscle fatigue and joint position sense in the shoulder that may play a role in impaired performance in repetitive overhead activities.[33]

Muscle Weakness Secondary to Neuropathy

Muscle weakness may be related to nerve involvement. Long thoracic nerve palsy has been identified as a cause of faulty scapular mechanics resulting from serratus anterior muscle weakness, leading to impingement in the suprahumeral region.[163]

Hypomobile Posterior GH Joint Capsule

Tightness in the posterior GH joint capsule compromises the normal arthrokinematics and increases forces on the head of the humerus against the anterior capsule. Harryman et al.[69] demonstrated increased anterior translation in the humeral head when there is a tight posterior capsule.

Summary of Common Impairments with Rotator Cuff Disease and Impingement Syndromes

Some, all, or none of the following may be present.

- Pain at the musculotendinous junction of the involved muscle with palpation, with resisted muscle contraction, and when stretched
- Positive impingement sign (forced internal rotation at 90° of flexion) and painful arc
- Impaired posture: thoracic kyphosis, forward head, and forward (anterior) tipped scapula with decreased thoracic mobility
- Muscle imbalances: hypomobile pectoralis major and minor, levator scapulae, and internal rotators of the GH joint; weak serratus anterior and lateral rotators
- Hypomobile posterior GH joint capsule
- Faulty kinematics with humeral elevation: decreased posterior tipping of scapula related to weak serratus anterior; scapular elevation and overuse of upper trapezius; and uncoordinated scapulohumeral rhythm

- With a complete rotator cuff tear, inability to abduct the humerus against gravity
- When acute, pain referred to the C5 and C6 reference zones

Common Functional Limitations/Disabilities

- When acute, pain may interfere with sleep, particularly when rolling onto the involved shoulder.
- Pain with overhead reaching, pushing, or pulling.
- Difficulty lifting loads.
- Inability to sustain repetitive shoulder activities (such as reaching, lifting, throwing, pushing, pulling, or swinging the arm).
- Difficulty with dressing, particularly putting a shirt on over the head.

Management: Painful Shoulder Syndromes (Without Dislocation)

NOTE: Even though symptoms may be "chronic" in terms of long standing or recurring, if there is inflammation the initial treatment priority is to get the inflammation under control.

Management: Protection Phase

Control Inflammation and Promote Healing

- Modalities and low-intensity cross-fiber massage are applied to the site of the lesion. While applying the modalities, position the extremity to maximally expose the involved region.[42,45]
- Support the arm in a sling for rest.

Patient Education

The environment and habits that provoke the symptoms must be modified or avoided completely during this stage. The patient should be informed about the mechanics of the irritation and given guidelines for anticipated recovery with compliance.

Maintain Integrity and Mobility of the Soft Tissues

- Passive, active-assistive, or self-assisted ROM is initiated in pain-free ranges.
- Multiple-angle muscle setting and protected stabilization exercises are initiated. Of particular importance in the shoulder is to stimulate the stabilizing function of the rotator cuff, biceps brachii, and scapular muscles at an intensity tolerated by the patient.

PRECAUTION: It is important to use caution with exercises during this stage to avoid the impingement positions. Often the mid-range of abduction, scaption with internal rotation, or an end-range position when the involved muscle is on a stretch (such as putting the hand behind the back) provokes a painful response.

Control Pain and Maintain Joint Integrity

Pendulum exercises without weights can be used to cause pain-inhibiting grade II joint distraction and oscillation motions (see Fig. 17.22 in the section on exercise).

Develop Support in Related Regions

- Postural awareness and correction techniques are used. (See related information on interventions for impaired posture in Chapter 14.)
- Supportive techniques such as shoulder strapping or scapular taping, tactile cues, and mirrors can be used for reinforcement. Repetitive reminders and practice of correct posture is necessary throughout the day.

Management: Controlled Motion Phase

Once the acute symptoms are under control, the main emphasis becomes use of the involved region with progressive, nondestructive movement and proper mechanics while the tissues heal. The components of the desired functions are analyzed and initiated in a controlled exercise program.[44,45,90,198,199] If there is a functional laxity in the joint, the intervention is directed toward learning neuromuscular control of and developing strength in the stabilizing muscles of both the scapula and glenohumeral joint.[29,90,93,162,181] If there is restricted mobility that prevents normal mechanics or interferes with function, mobilization of the restricted tissue is performed. Exercise techniques and progressions are described later in the chapter.

Patient Education

Patient adherence with the program and avoidance of irritating the healing tissues are necessary. The home exercise program is progressed as the patient learns safe and effective execution of each exercise.

Develop a Strong, Mobile Scar

- Manual therapy techniques such as cross-fiber or friction massage are used. The extremity is positioned so the tissue is on a stretch if it is a tendon or in the shortened position if it is in the muscle belly. The technique is applied to the tolerance of the patient.
- Following massage, the patient is instructed to perform an isometric contraction of the muscle in several positions of the range. The intensity of contraction should not cause pain.
- The patient should be taught how to self-administer the massage and isometric techniques.

Improve Postural Awareness

It is important to continue to reinforce proper postural habits. Tactile and verbal cues are used to increase patient awareness of scapular and cervical posture every time an exercise is performed, such as touching the scapular adductors and chin and reminding the patient to "pull the shoulders back" and "lift the head" while doing the shoulder exercises.

Focus on Evidence

In a random placebo-controlled, crossover study of 120 subjects (60 with impingement and 60 without symptoms), changing posture resulted in a significant increase in ROM in flexion, abduction, and scaption; and the point in the range where symptoms were felt was significantly higher.

Thoracic and scapular taping had a positive influence in modifying posture; there was less forward head posture, smaller kyphosis, less lateral scapular displacement, less elevated and forward scapula position, and increased pain-free range of scaption compared with the measurements taken after placebo taping in both the symptomatic and asymptomatic groups.[99]

Modify Joint Tracking and Mobility

Mobilization with movement (MWM) may be useful for modifying joint tracking and reinforcing full movement when there is painful restriction of shoulder elevation because of a painful arc or impingement[116] (see Chapter 5 for a description of principles).

- Posterolateral glide with active elevation (Fig. 17.17A)
  Patient position: Sitting with the arm by the side and head in neutral retraction.
  Therapist position and procedure: Stand on the side opposite the affected arm and reach across the patient's torso to stabilize the scapula with the palm of one hand. The other hand is placed over the anteromedial aspect of the head of the humerus. Apply a graded posterolateral glide of the humeral head on the glenoid. Request that the patient perform the previously painful elevation. Maintain the posterolateral glide mobilization throughout both elevation and return to neutral. Ensure that no pain is experienced during the procedure. Adjust the grade and direction of the glide as needed to achieve pain-free function. Add resistance in the form of elastic resistance or a cuff weight to load the muscle.

Self-Treatment. A mobilization belt provides the posterolateral glide while the patient actively elevates the affected limb against progressive resistance to end range (Fig. 17.17B).

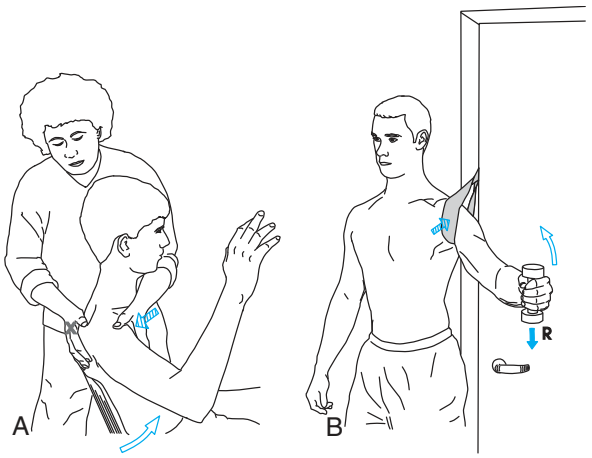


FIGURE 17.17 MWM to modify joint tracking and improve active elevation. A posterolateral glide is applied to the humeral head (*A*) manually or (*B*) with a belt for self-treatment, while the patient actively elevates the humerus. A weight is used to strengthen the muscles through the pain-free range.

Develop Balance in Length and Strength of Shoulder Girdle Muscles

It is important to design a program that specifically addresses the patient's impairments. Typical interventions in the shoulder girdle include but are not limited to:

- *Stretch shortened muscles.* Shortened muscles typically include the pectoralis major, pectoralis minor, latissimus dorsi and teres major, subscapularis, and levator scapulae.
- *Strengthen and train the scapular stabilizers.* Scapular stabilizers typically include the serratus anterior and lower trapezius for posterior tipping and upward rotation and the middle trapezius and rhomboids for scapular retraction. It is important that the patient learns to avoid scapular elevation when raising the arm. Therefore, practice scapular depression when abducting and flexing the humerus.
- *Strengthen and train the rotator cuff muscles*, especially the shoulder lateral rotators.

Develop Muscular Stabilization and Endurance

- Alternating isometric resistance is applied to the scapular muscles in *open-chain positions* (side-lying, sitting, supine), including protraction/retraction, elevation/depression, and upward/downward rotation so the patient learns to stabilize the scapula against the outside forces (see Fig. 17.36 and the exercise section).
- Scapular and glenohumeral patterns are combined using flexion, abduction, and rotation. Alternating isometric resistance is applied to the humerus while the patient holds against the changing directions of the resistance force (see Figs. 17.37, 17.38, and 17.41 in the exercise section).
- *Closed-chain stabilization* is performed with the patient's hands fixated against a wall, a table, or the floor (quadruped position) while the therapist provides a graded, alternating isometric resistance or rhythmic stabilization. Observe for abnormal scapular winging. If it occurs, the scapular stabilizers are not strong enough for the demand; so the position should be changed to reduce the amount of body weight (see Fig. 17.42 in the exercise section).
- *Muscular endurance* is progressed by increasing the amount of time the individual holds the pattern against the alternating resistance. The limit is reached when any one of the muscles in the pattern can no longer maintain the desired hold. The goal at this phase should be stabilization for approximately 3 minutes.

Progress Shoulder Function

As the patient develops strength in the weakened muscles, it becomes important to develop a balance in strength of all shoulder and scapular muscles within the range and tolerance of each muscle. To increase coordination between scapular and arm motions, dynamically load the upper extremity within tolerance of the synergy with submaximal resistance. To improve muscular endurance, have the patient increase control from 1 minute to 3 minutes.

BOX 17.8 Patient Instructions to Prevent Recurrences of Shoulder Pain

- Prior to exercise or work, massage the involved tendon or muscle; follow with isometric resistance and then with full ROM and stretching of the muscle.
- Take breaks from the activity if repetitive in nature. If possible, alternate the stressful, provoking activity with other activities or patterns of motion.
- Maintain good postural alignment; adapt seating or workstation to minimize stress. If sport-related, seek coaching in proper techniques or adapt equipment for safe mechanics.
- Prior to initiating a new activity or returning to an activity for which not conditioned, begin a strengthening and training program.

Management: Return to Function Phase

Specificity of training toward the desired functional outcome begins as soon as the patient has developed control of posture and the basic components of the desired activities without exacerbating the symptoms. While working with the patient, continue to instruct him or her on how to progress the program when discharged as well as how to prevent recurrences. Suggestions are summarized in Box 17.8.

Increase Muscular Endurance

To increase muscular endurance, repetitive loading of the defined patterns is increased from 3 minutes to 5 minutes.

Develop Quick Motor Responses to Imposed Stresses

- The stabilization exercises are applied with increased speed.
- Plyometric training in both open-chain and closed-chain patterns is initiated if power is a desired outcome.

Progress Functional Training

Specificity of training progresses to an emphasis on timing and sequencing of events.

- Eccentric training is progressed to maximum load.
- Desired functional activities are simulated—first under controlled conditions, then under progressively challenged situations using acceleration/deceleration drills.
- The patient is involved in assessing performance in terms of safety, symptom provocation, postural control, and ease of execution and then practices adaptations to correct any problems.

PAINFUL SHOULDER SYNDROMES: SURGERY AND POSTOPERATIVE MANAGEMENT

Surgical intervention is an option for painful shoulder syndromes when conservative management does not resolve symptoms and improve a patient's function. For an individual with primary impingement as a result of structural variations in the acromion (see descriptions and Fig. 17.16 in the previous section), subacromial decompression may be performed. An individual with a partial- or full-thickness rotator cuff tear may require surgical repair. The subacromial decompression and rotator cuff repair surgeries and postoperative management are described in this section. Surgical repair and postoperative management due to instabilities and dislocations are described in a later section of this chapter.

Subacromial Decompression and Postoperative Management

When pain and loss of functional mobility associated with primary impingement do not resolve sufficiently with nonoperative management, *subacromial decompression,* designed to increase the volume of the subacromial space and provide adequate gliding room for tendons, followed by postoperative rehabilitation is often warranted. Subacromial decompression, is commonly referred to as *anterior acromioplasty* or *decompression acromioplasty.* However, acromioplasty, which alters the shape of the acromion, is typically but not always one of the components of subacromial decompression.[111]

Indications for Surgery

The following are generally accepted indications for surgical management of impingement syndromes.*

- Pain during overhead activities and loss of functional mobility of the shoulder as the result of primary impingement that persists (typically for 3 to 6 months or longer) despite a trial of nonoperative interventions.
- Stage II (Neer classification) (see Box 17.7) impingement with nonreversible fibrosis or bony alterations of the subacromial compartment, calcific deposits in the cuff tendons, and symptomatic subacromial crepitus.
- Intact or minor tear of the rotator cuff.

NOTE: Patients who present with secondary impingement (GH joint hypermobility or instability associated with a partial- or full-thickness tear of the rotator cuff) are not candidates for surgical subacromial decompression alone. For these patients subacromial decompression is combined with concomitant repair of the cuff tear; otherwise the procedures inherent in subacromial decompression can worsen GH instability.[71,111,190]

Procedures

Surgical approach. Subacromial decompression is performed using an arthroscopic or open approach. Although an open approach has been used successfully for many years,[75,117,120,149] the preferred procedure today in most cases is an arthroscopic approach.[53,190] Unlike a traditional open approach, in which the proximal attachment of the deltoid must be detached and then repaired prior to

*See references 1, 53, 71, 75, 111, 120, 149, 155, 190.

closure,[120] with an arthroscopic approach the deltoid remains intact, enabling the patient to regain functional use of the upper extremity more rapidly after surgery.

Another option, preferred by some surgeons to either a traditional open or arthroscopic decompression, is a "mini-open" approach, which involves splitting the deltoid insertion vertically rather than detaching it.[111] For the most part, the traditional open approach for subacromial decompression is now reserved for some patients with a massive rotator cuff tear who also are undergoing an open repair.

Component procedures. There are several surgical procedures that can be performed for subacromial decompression depending on the pathology observed during examination of the shoulder prior to or during surgery.[1,53,67,71,111,131,155,190]

- Removal of the subacromial bursa (*bursectomy*), which is typically thickened (enlarged) by chronic inflammation.
- Release of the coricoacromial ligament, which is usually hypertrophied and may also be frayed, followed by complete or partial resection or recession.
- Resection of the anterior acromial protuberance and contouring the undersurface of the remaining portion of the acromion (*acromioplasty*) to enlarge the subacromial space (Fig. 17.18).
- Removal of any osteophytes at the AC joint and in some cases resection of the distal portion of the clavicle for advanced arthritis of the AC joint.

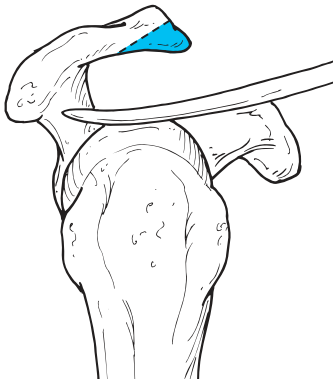


FIGURE 17.18 Arthroscopic acromioplasty showing the line of resection of the anterior acromion.

Postoperative Management

The type of surgical approach used and the status of the rotator cuff significantly affect the progression of rehabilitation after subacromial decompression (acromioplasty). More often than not an arthrocopic approach is used. If the rotator cuff is intact preoperatively, rehabilitation after arthroscopic acromioplasty progresses quite rapidly because the shoulder musculature is left intact during the procedure. In contrast, if a repair of the rotator cuff is required in addition to acromioplasty or if the surgeon elects to use a mini-open or open approach, rehabilitation progresses at a slower rate to allow the repaired shoulder musculature adequate time to heal.

NOTE: The guidelines outlined in this section are for postoperative rehabilitation after *arthroscopic* subacromial decompression for a patient with primary shoulder impingement who has an *intact rotator cuff*. If subacromial decompression is combined with repair of the rotator cuff (arthroscopic, mini-open, or open approach), the guidelines presented in a later section of this chapter on rehabilitation after rotator cuff repair are appropriate.

Immobilization

The shoulder is immobilized and supported in a sling with the arm positioned at the patient's side or in slight abduction, the shoulder is internally rotated, and the elbow is flexed to 90°. The sling is worn for comfort for 1 to 2 weeks but is removed for exercise the day after surgery.[111,190,200]

Exercise

Exercise interventions after subacromial decompression target the same impairments common to rotator cuff impingement as do the exercises selected for a nonoperative program. These impairments and the associated pathomechanics of the shoulder girdle were discussed and summarized previously in this chapter. This information merits review to understand why specific exercises are included in the following postoperative rehabilitation program.

The suggested criteria and timelines for a patient to progress from one phase of rehabilitation to the next, as well as the goals, exercises, and activities in this section, are drawn from several published resources.[3,37,71,91,111,190,200] The exercises noted for each phase impose gradually increasing demands as soft tissues heal. In addition, patient education, emphasizing *prevention* of postures and movements that contribute to impingement and provoke pain, is woven throughout each phase of rehabilitation.

Because arthroscopic decompression often is performed on an outpatent basis, a patient must carry out the prescribed exercises at home with little supervision. Therefore, on the few visits the patient may have with a therapist, patient education is of the utmost importance for each phase of rehabilitation.

Exercise: Maximum Protection Phase

The first phase of rehabilitation after arthroscopic decompression begins on the day after surgery and extends for 3 to 4 weeks. Emphasis is placed on pain control and immediate but comfortable assisted movement of the shoulder to prevent adhesions of the cuff tendons in the subacromial space. Attaining full or nearly full passive ROM of the operated shoulder (compared to the noninvolved shoulder) is a reasonable goal by 4 to 6 weeks postoperatively.[71]

Patient education begins immediately and is directed toward helping the patient recognize and prevent postures during exercise and ADL that contribute to symptoms. Active (unassisted) shoulder ROM is permissible as soon as motions are pain-free and proper scapulothoracic and glenohumeral control can be maintained. This may be possible as early as 2 weeks after arthroscopic decompression

if the rotator cuff did not require repair and is functioning sufficiently.

Goals and interventions. The following goals and exercises are indicated for the early stage of tissue healing.[1,3,37,71,111,200]

- ***Control pain and inflammation.***
  - Use of a sling when the arm is dependent.
  - Use of cryotherapy and prescribed anti-inflammatory medication.
  - Active ROM of the cervical spine and shoulder relaxation exercises.
- ***Prevent loss of mobility of peripheral joints.***
  - Active ROM of the elbow, wrist, and hand.
- ***Develop postural awareness and control.***
  - Active exercises of the scapula with emphasis on retraction.
  - Posture training, reinforcing an erect trunk (avoiding excessive thoracic kyphosis) for prevention of postures that can contribute to impingement symptoms during exercises and ADL.
- ***Restore pain-free shoulder mobility.***
  - Pendulum exercises for pain control and mobility.
  - Assisted shoulder ROM as tolerated by pain, initially guiding with the sound upper extremity and later a wand. Start in the supine position to provide additional stability to the scapula against the thorax and with the upper arm on a folded towel in slight abduction and flexion. Shoulder motions include elevating the arm in the plane of the scapula, forward flexion, abduction, rotation, and horizontal abduction and adduction. Progress to performing exercises in a semi-reclining position and then in a seated or standing position while maintaining thoracic extension.
  - Assisted shoulder extension in a standing position with a wand held behind the back.
  - Stretching the posterior shoulder structures in pain-free range using a cross-chest stretch into horizontal adduction. Postpone until next phase if painful.
  - Active ROM (unassisted) of the shoulder and scapula within pain-free ranges maintaining proper scapulothoracic and glenohumeral control; begin supine and progress to sitting. Active shoulder motions may be possible by 2 weeks postoperatively.
- ***Prevent reflex inhibition and atrophy of shoulder girdle musculature.***
  - *Pain-free,* low-intensity, multiple-angle isometrics of GH musculature with the arm supported and emphasis on the rotator cuff against minimal resistance. Begin submaximal isometrics a week or so postoperatively. Lightly resist with the uninvolved upper extremity. Focus on increasing repetitions more than resistance.[103,163]
  - Submaximal alternating isometric and rhythmic stabilization exercises for scapulothoracic muscles with the involved arm supported by the therapist. Target the scapular retractors and upward rotators.

Exercise: Moderate Protection Phase

Exercises during the second phase of rehabilitation are directed toward attaining full, pain-free shoulder ROM and improving neuromuscular control and muscle performance (strength, muscular endurance) of the deltoid, rotator cuff, and scapular stabilizers. The patient may be ready to begin this phase of rehabilitation as early as 3 to 4 weeks postoperatively but more often by 4 to 6 weeks. This phase extends over a 4- to 6-week period or until the patient meets the criteria to progress to the next phase.

Criteria for progression. Criteria to advance to the second phase include[37,71,91,111,200]:

- Minimal discomfort when the shoulder is unsupported; arm swing is comfortable during ambulation.
- Nearly complete, pain-free, *passive* ROM of the shoulder (full mobility of the scapula; at least 150° of shoulder elevation; full rotation).
- In the supine position, pain-free *active* elevation of the arm well above the level of the shoulder.
- Pain-free, *active* external rotation of the shoulder to about 45°.
- At least fair (3/5) and preferably good (4/5) muscle testing grade of shoulder musculature.

Goals and interventions. The goals, exercises, and activities during the second (intermediate) phase of rehabilitation are[37,71,91,200]:

- ***Restore and maintain full, pain-free passive mobility of the shoulder girdle and upper trunk.***
  - Joint mobilization, emphasizing posterior and caudal glides of the humerus and scapulothoracic mobility.
  - Low-intensity self-stretching of range-limiting muscles that could restrict sufficient upward rotation of the scapula and rotation of the humerus necessary for full elevation of the arm overhead, specifically the levator scapulae, rhomboids, middle trapezius, latissimus dorsi, and pectoralis major and minor. Remember, shortening of these muscles contributes to impingement of soft tissues during overhead movements of the arm.
  - Self-stretching (cross-chest stretch) of the posterior deltoid and posterior capsule of the GH joint. These structures are usually tight in the presence of shoulder impingement. A tight posterior capsule causes excessive anterior translation and superior migration of the head of the humerus in the glenoid, which in turn causes impingement of soft tissues during overhead reach.[69]
  - Self-stretch of the upper trunk to increase thoracic extension by lying supine on a rolled towel placed vertically between the scapulae.
  - Use the gained ROM in exercises and functional movement patterns during ADL.
- ***Reinforce posture awareness and control.***
  - Through patient education, continue to incorporate an erect trunk into exercises and function activities.

- ***Develop dynamic stability, strength, endurance, and control of scapulothoracic and GH muscles.***
  - Stabilization exercises against increasing resistance and in weight-bearing postures. If winging of the scapula occurs with weight bearing, provide manual support or decrease the imposed loads. Emphasize isolated strengthening of the serratus anterior and trapezius muscles.
  - Upper extremity ergometry for muscular endurance. To avoid an impingement arc, initiate in a standing position rather than while seated.
  - Dynamic strengthening exercises of isolated shoulder muscles against low loads (1- to 5-lb weight or light-grade elastic tubing), gradually increasing repetitions. Begin resisted elevation of the arm in the supine position to stabilize the scapula against the thorax; progress to sitting or standing.
  - Use the involved arm for functional activities that involve light resistance.

NOTE: Target the upward rotators of the scapulothoracic joint (serratus anterior, upper and lower trapezius) and the rotator cuff muscles,[166] as well as the latissimus dorsi, teres major, and biceps brachii, which act as humeral head depressors and therefore oppose superior translation during active elevation of the arm. Initially perform resisted motions of the humerus below the level of the shoulder; later progress to overhead exercises if motions remain pain-free.

PRECAUTION: Be certain the patient can perform active shoulder flexion and abduction against gravity without hiking the shoulder before progressing to resisted exercises above shoulder level.

Exercise: Minimum Protection/ Return to Function Phase

The final phase of rehabilitation usually begins by 8 weeks postoperatively, at which time soft tissues are reasonably well healed and require little to no protection. Exercises continue until about 12 to 16 weeks postoperatively or until the patient has returned to full activity. Exercises are directed toward continuing to improve strength and endurance of the shoulder girdle muscles using isolated movements and those that simulate functional activities. Patients often see continued improvement in functional use of the operated upper extremity for 6 months postoperatively.[3]

The time necessary for full recovery and unrestricted activities depends largely on the level of demand of the anticipated activities. A patient wishing to return to competitive sports requires a more demanding progression of advanced exercises (e.g., plyometric training and sport-specific drills) than a sedentary individual.[37,200,203]

Criteria for Progression. The criteria that should be met to progress to the final phase of rehabilitation are[37,91,200]:

- Negative impingement test.
- Full, pain-free, active ROM of the shoulder without evidence of substitute motions.
- Seventy-five percent or more strength of the shoulder musculature compared with the sound shoulder.[200]

Goals and Interventions. The goals, exercises, and activities during the final phase of rehabilitation after subacromial decompression and for the final phase of nonoperative management of primary impingement syndrome are similar. Refer to the information presented in the previous section of this chapter as well as other resources.[37,45,113,198,200,203]

Outcomes

There appears to be no significant difference in the *long-term* results (pain-free ROM and return to desired functional activities) after either open or arthroscopic surgery for primary impingement syndrome with or without associated rotator cuff disease.[53,111,190] Based on the results of numerous outcome studies of open and arthroscopic procedures, 85% to 95% of patients report good to excellent results 1.0 to 2.5 years postoperatively.[1,71,111,190] In general, patients reporting the least satisfaction with their function after surgery are those who participate in high-demand athletic activities that involve overhead throwing and those with work-related injuries who are receiving workers' compensation.[111]

Follow-up studies have documented several advantages of an arthroscopic over comparable open surgical management of impingement syndrome. They include less postoperative pain; earlier restoration of full ROM and strength; earlier return to work (often as early as 1 week postoperatively); lower cost (shorter hospital stay or outpatient surgery); and a more favorable cosmetic result.[1,71,111,190]

Although exercises are routinely prescribed after subacromial decompression, the effectiveness of exercise has been the focus of very few studies. One prospective, randomized study carried out in Denmark looked at the effectiveness of a 6-week therapist-supervised exercise program compared to a self-managed program after arthroscopic subacromial decompression.[3] Patients in the therapist-supervised group received exercise instruction while in the hospital and then for a 1-hour therapy session once a week for 6 weeks after discharge from the hospital. Patients in the self-managed group received exercise instruction on one occasion prior to discharge from the hospital. Both groups received written instructions. At 6 weeks and 3, 6, and 12 months, all patients demonstrated improvement in all parameters tested. However, there were no significant diffferences between the two groups with the exception of one measurement. At 3 months postoperatively, the therapist-supervised group had a higher level of pain than the self-managed group. The authors concluded that initial, therapist-directed exercise instruction followed by a home-based, self-managed exercise program achieved rehabilitation goals as effectively as a therapist-supervised program.

Rotator Cuff Repair and Postoperative Management

There are two broad categories of rotator cuff tears, defined by the depth of the tendon tear: partial-thickness and full-thickness tears. Either type may require surgical

management. A *partial-thickness tear* extends inferiorly or superiorly through only a portion of the tendon from either the acromial (bursal) or humeral (articular) surface of the tendon. A *full-thickness tear* is a complete tear, which extends the entire depth of the tendon.[71,77,111]

Indications for Surgery

The primary indications for surgical management of a rotator cuff tear confirmed by imaging are pain and impaired function as the result of the following.*

- Partial-thickness or full-thickness tears of the rotator cuff tendons resulting from repetitive microtrauma and chronic impingement, which lead to irreversible degenerative changes in soft tissues. Some patients with stage II lesions and most with stage III lesions (Neer classification[118]) who continue to be symptomatic and have functional limitations after a trial of nonoperative treatment are candidates for surgery.
- Acute, traumatic rupture (frank, full-thickness tear) of the rotator cuff tendons often combined with avulsion of the greater tuberosity, labral damage, or acute dislocation of the GH joint in individuals with no known history of cuff injury. Full-thickness, traumatic tears occur most often in young, active adults.

NOTE: Surgical repair is not indicated in patients who are asymptomatic despite the presence of a cuff tear confirmed by imaging.

Procedures

Depending on the severity and location of a tear of one or more of the rotator cuff tendons, the extent of associated lesions, the onset of the tear (repetitive microtrauma or traumatic injury), the quality and mobility of the torn tissues, bone quality, patient-related considerations (age, health, activity level) and the surgeon's preference and experience, there are several operative options for repairing a torn rotator cuff.[71,72,111,190]

Type of Repair

The type of cuff repair is typically classified by the surgical approach and techniques used. There are three categories of repair.

- ***Arthroscopic approach.*** The entire procedure is performed arthroscopically and requires only a few small incisions for port sites.[8,59,60,71,111,172,190,210]
- ***Mini-open (arthroscopically assisted) approach.*** There are two variations of this type of procedure, both of which involve arthroscopic subacromial decompression and a *deltoid-splitting* approach. In one variation only the subacromial decompression is performed arthroscopically, whereas in the other variation a portion of the cuff repair itself is also performed arthroscopically.[210] In both cases, an anterolateral incision is made at the acromion and is extended distally (either 1.5 or 3.5 cm but no more than 4 cm to avoid the axillary nerve) along the fibers of the deltoid insertion. The deltoid is split longitudinally between its anterior and middle portions to allow visualization of the cuff tear without detaching the deltoid from its proximal insertion.[56,111,138,180]
- ***Traditional open approach.*** An anterolateral incision is made that extends obliquely from the middle one-third of the inferior aspect of the clavicle, across the coracoid process, and to the anterior aspect of the proximal humerus. The proximal insertion of the deltoid must be detached and reflected for exposure of the operative field during an open subacromial decompression and cuff repair. After the cuff repair is complete, the deltoid is reattached to the acromion.† As arthroscopic and arthroscopically assisted repairs of the rototor cuff have advanced, the use of the traditional open approach has decreased.

Components of a Rotator Cuff Repair

Regardless of the approach, subacromial decompression is performed (particularly for cuff tears associated with chronic impingement) before repair of the cuff is undertaken. After the tear is visualized through the approach selected, the margins of the torn tendon are débrided. The tendon is then released from any adherent soft tissues and mobilized for advancement and apposition to bone that has been prepared for sutures, followed by *tendon-to-bone* fixation. Depending on whether an arthroscopic or mini-open approach is used, fixation is accomplished by sutures and suture anchors, tacks, or staples.[56,59,71,111,179,190]

In addition to subacromial decompression, other concomitant procedures may be required.[71,111] For example, capsular tightening or labral reconstruction may be performed if unidirectional or multidirectional instability of the GH joint is present. Because degenerative changes in the tendon of the long head of the biceps brachii are often associated with chronic rotator cuff disease, a repair of this tendon may also be necessary.

Selection of Surgical Procedures

The surgeon weighs many factors when determining which type of cuff repair is most appropriate for each patient. One such consideration is the severity of the tear, including thickness (partial or full), size, and number of tendons torn. Although there is some variability in the literature, there are four generally accepted categories that describe the longitudinal size of rotator cuff tears: *small* (1 cm or less), *medium* (1 to 3 cm), *large* (3 to 5 cm), and *massive* (more than 5 cm or a full-thickness tear of more than one tendon).[8,56,71,191]

If a small, *partial-thickness* cuff tear is managed surgically after failed nonoperative treatment, a fully arthroscopic approach typically is performed to débride the frayed margins of the torn tendon, coupled with a subacromial decompression procedure. However, the torn portion of the tendon may or may not be repaired.[8,71,111,172,179,190]

Two decades ago, primarily small and medium *full-thickness* cuff tears (≤ 1 to 3 cm) were managed with a fully arthroscopic approach.[60,71,111] With the evolution of arthroscopic techniques, an increasing number of large,

*See references 8, 12, 71, 77, 111, 138, 140, 167, 178, 190, 210.

†See references 12, 77, 97, 111, 140, 178.

full-thickness cuff tears and occasionally massive tears as well are managed with a fully arthroscopic approach.[190,210] However, variations of the mini-open (deltoid splitting) approach frequently are the surgeon's choice for repair of medium and large tears.[56,111,146,180] Even some massive tears (> 5 cm) are managed with a deltoid-splitting approach.[111,191] The traditional open approach, requiring detachment and repair of the deltoid, now primarily is reserved for repairs of multiple tendon tears associated with extensive injury to the shoulder or for use as a component of an open surgical procedure, such as total shoulder replacement arthroplasty.[56,111]

The location of the cuff tear, amount of retraction and mobility of a full-thickness tear, and the quality of the remaining tendon and underlying bone also influence the surgeon's selection of the type of cuff repair expected to be most effective.[71,111,179,190] Whereas small, medium, and large tears of the supraspinatus (the most frquently torn cuff tendon) or infraspinatus are routinely managed with arthroscopic or mini-open approaches, tears of the subscapularis are often managed with a traditional open approach.[56] If there is significant retraction and poor mobility of the torn tendon or poor tissue quality, many surgeons believe that a stronger repair can be achieved with an open procedure (mini-open or traditional) than with an arthroscopic repair.[71]

Postoperative Management

After surgical repair of a torn rotator cuff tendon, there are many factors that influence decisions about the position and duration of immobilization, the selection and application of exercises, and the rate of progression of each patient's postoperative rehabilitation program. These factors and their potential impact are summarized in Table 17.3. Furthermore, these factors can affect postoperative outcomes.

There is little consensus reported in the literature or practiced in the clinical setting as to how and to what extent each of these factors, singularly or collectively, has an impact on the decisions made by the surgeon and the therapist about a patient's postoperative rehabilitation program. Hence, predetermined guidelines and protocols for postoperative management after rotator cuff repair are diverse and sometimes contradictory.* For example, some authors have pointed out that if deltoid detachment and repair are components of the surgery, as is necessary for a traditional open repair, deltoid strengthening exercises should be postponed for approximately 6 to 8 weeks postoperatively until the repaired deltoid has healed.[37,52,111] Yet another author suggested that rehabilitation should proceed similarly regardless of whether deltoid detachment was required so long as secure fixation of the deltoid was achieved.[71]

*See references 8, 24, 37, 50, 52, 56, 71, 111, 138.

TABLE 17.3 Factors that Influence Progression of Rehabilitation After Repair of the Rotator Cuff

Factors	Potential Impact on Rehabilitation
• Onset of injury	• Chronic impingement and atraumatic cuff deficiency → slower progression than after acute traumatic injury.
• Size and location of the tear	• Larger tears with more structures involved and probability of more extensive surgery → slower progression.
• Associated pathologies such as GH instability or fracture	• Associated pathologies often lengthen the period of immobilization → slower progression of exercises or the need for additional precautions.
• Preoperative strength and mobility of the shoulder	• Preexisting weakness and atrophy of the dynamic stabilizers or limited passive and active mobility of the shoulder → slower postoperative progression.
• Patient's general health	• Patient in poor health; history of smoking; history of inflammatory disease → slower progression.
• History of steroid injections or previous, failed cuff surgery	• Compromised bone and tendon tissue quality, which affects the security of the repair (fixation) → slower progression.
• Preinjury level of activity of postoperative goals	• Higher level goals require a more extended and advanced postoperative training program because of a higher risk of reinjury.
• Age of patient	• Older patient who has an insidious (chronic) onset and may have articular changes → slower progression.
• Type of approach	• Traditional open approach (with deltoid detachment and repair) → slightly slower progression than after an arthroscopic or arthroscopically assisted (mini-open/deltoid splitting) repair.
• Type of repair	• Tendon to tendon → slower progression than tendon to bone.
• Mobility (no excessive tension on the repaired tendon when arm at side) and integrity of the repair	• If mobility is inadequate → longer duration of exercise within a protected ROM during early rehabilitation.
• Patient's compliance with the program	• Lack of compliance (doing too much or too little) can affect outcome.
• Philosophy, skill, and training of the surgeon	• All have an impact that could → either slower or more accelerated progression.

Given the diverse characteristics of patients undergoing repair of the rotator cuff and the variety of surgical options available, it is not surprising that no single postoperative program can be used for all patients or has been shown to yield better outcomes than another. Therefore, to meet each patient's needs and goals, a therapist can use published protocols or those developed at individual clinical facilities as general guidelines for postoperative management and can modify the guidelines based on ongoing examination of the patient's response to interventions and close communication with the surgeon.

Despite variations among postoperative programs, they share three common elements: (1) immediate or early postoperative motion of the GH joint; (2) control of the rotator cuff for dynamic stability; and (3) gradual restoration of strength and muscular endurance. The purpose of this section is to present *general* exercise guidelines that incorporate these elements into progressive phases of rehabilitation after arthroscopic or mini-open repair of a *full-thickness* cuff tear and to identify the differences in precautions and the progression of exercises for a traditional open procedure or differences based on the size and location of the tear and the quality of the repair.

NOTE: The goals, exercise interventions, and progression of rehabilitation after débridement rather than repair of a *partial-thickness* tear are comparable to postoperative management after subacromial decompression for cuff impingement presented in the previous section of this chapter.

Immobilization

The position and duration of immobilization of the operated shoulder after rotator cuff repair are dependent on many factors, such as the severity and location of the tear and the type and quality of the repair. One of those factors, the size of the cuff tear, in part determines whether the patient's operated arm is supported in a sling (shoulder adducted, internally rotated, and elbow flexed to 90°) or in an abduction orthosis or pillow (shoulder elevated in the plane of the scapula approximately 45°, shoulder internally rotated, and elbow flexed). Some patients, using an abduction splint, initially may be required to have a family member support the operated arm in the 45° shoulder position when the splint is removed for exercise or dressing and even for bathing.

Table 17.4 summarizes the variations noted in the literature for fully arthroscopic and mini-open/deltoid-splitting approaches. Immobilization after a traditional open procedure that involves deltoid detachment and repair, is not included in Table 17.4 because of the variations in guidelines reported in the literature.[37,71,111,191]

The rationale for initially immobilizing the operated shoulder in abduction is based on two principles. (1) In the abducted position, the repaired cuff, as well as the reattached deltoid with an open approach, are held in a relaxed position, which reduces the possibility of reflexive muscle contractions that could disrupt the repairs. (2) Supporting the arm in abduction rather than adduction reduces tension on the tendons and therefore may improve blood flow to the repaired tendon(s).

Exercise

Regardless of whether a patient undergoes a rotator cuff repair on an inpatient or outpatient basis, contact with a therapist for exercise instruction after surgery is usually limited to a few visits unless the patient does not progress satisfactorily. Therefore, the emphasis of a therapist's interaction with a patient must be placed on patient education for an effective and safe home-based exercise program. As with rehabilitation after arthroscopic subacromial decompression,[3] the extent of therapist-directed exercise after rotator cuff surgery also has been investigated.

Focus on Evidence

In a randomized, controlled study by Roddey and colleagues,[150] two approaches to exercise instruction following arthroscopic repair of a full-thickness rotator cuff tear were compared, specifically in-person instruction by a therapist and video-based instruction. On the first postoperative day, both groups of patients (total 108) received one visit from the therapist for initial instruction in the postoperative program (sling use and passive shoulder exercises). Patients in both groups received written handouts about the home exercise program. In addition, patients in the video-instruction group received a video containing exercises for all phases of the rehabilitation program.

After discharge, patients in the video group saw the therapist four times (at 2, 6, 12, and 24 weeks) for evalua-

TABLE 17.4 Relationships of Type and Duration of Immobilization after Arthroscopic and Mini-Open Repair* to the Size of the Rotator Cuff Tear**

Size of Tear	Type and Duration of Immobilization
Small (≤ 1 cm)	Sling for 1–2 weeks; removal for exercise the day of surgery or 1 day postop
Medium to large (> 5 cm)†	Sling or abduction orthosis/pillow for 3–6 weeks; removal for exercise 1–2 days postop
Massive (> 5 cm)†	Sling or abduction orthosis/pillow for 4–8 weeks; removal for exercise 1–3 days postop

*Fully arthroscopic and mini-open (arthroscopically assisted//deltoid splitting) approaches.

**Age of the patient, onset of the injury, and quality of repaired tissue also influence the position and duration of immobilization.

†A fully arthroscopic approach is not often used to repair massive cuff tears.

tion and an okay to advance to the next phase of rehabilitation, but they received all exercise instruction by watching the video at home. Patients in the other group also saw the therapist four times at identical intervals after discharge for follow-up evaluations and one-to-one instructions from the therapist on how to perform the exercises during the next phase of the home program. Between visits both groups had telephone access to their therapist for questions, and at 52 weeks all patients were evaluated a final time.

Results of this study indicated that there were no significant differences between the two groups in compliance with the exercise program and functional outomes measured with a self-report instrument. The authors concluded that video-based exercise instruction was equally effective as therapist-directed exercise instruction. It is important to note that 30% of the patients dropped out of the study. The authors did not report whether these patients were progresssing well or if any of them left the study to seek individualized or more frequent therapy.

Goals and interventions for each phase of rehabilitation after arthroscopic or mini-open cuff repair follow. General guidelines for exercise and precautions after rotator cuff repair are summarized in Box 17.9. Precautions specific to a particular type of cuff tear or surgical procedure are also noted. The suggested timelines for each phase are general and must be adjusted based on factors already noted (see Table 17.3).

BOX 17.9 General Exercise Guidelines and Precautions After Repair of a Full-Thickness Rotator Cuff Tear

Early Shoulder Motion

- Perform passive or assisted shoulder ROM within *safe and pain-free ranges* based on the surgeon's intraoperative observation of the mobility and strength of the repair and the patient's comfort level during exercise.
- Only passive and no assisted ROM for 6 to 8 weeks after repair of a massive cuff tear or after a traditional open approach to prevent avulsion of the repaired deltoid.
- Initially perform passive and assisted shoulder ROM in the supine position to maintain stability of the scapula on the thorax.
- Minimize anterior translation of the humeral head and the potential for impingement. See that the humerus is positioned slightly anterior to the frontal plane of the body and in slight abduction
  - While at rest in the supine position support the humerus on a folded towel.
  - When the initiating passive or assisted shoulder rotation while lying supine, position the shoulder in slight flexion and approximately 45° of abduction.
  - When initiating assisted shoulder extension, perform the exercise in prone (arm over the edge of the bed) from 90° to just short of neutral. Later progress to exercises behind the back.
- When performing assisted or active exercises in the upright position (sitting or standing), be certain that the patient maintains an erect trunk posture to minimize the possibility of impingement.
- To ensure adequate humeral depression and avoid superior translation of the head of the humerus when beginning *active* elevation of the arm, restore strength in the rotator cuff, especially the supraspinatus and infraspinatus muscles, before dynamically strengthening the shoulder flexors and abductors.
- Do not allow *active* shoulder flexion or abduction until the patient can lift the arm without hiking the shoulder.

Strengthening Exercises

- When beginning isometric resistance to scapulothoracic musculature, be sure to support the operated arm to avoid excessive tension in repaired GH musculature.
- Use low exercise loads; resisted motions should not cause pain.
- No weight-bearing (closed chain) exercises or activities for 6 weeks.
- Delay dynamic strengthening (progressive resistive exercise, or PRE) for a *minimum* of 8 weeks postoperatively for small, strong repair and for at least 3 months for larger tears.
  - If the supraspinatus was repaired, proceed cautiously when resisting external rotation.
  - If the subscapularis was repaired, proceed cautiously with resisted internal rotation.
- After an open repair, postpone isometric resistance exercises to the repaired deltoid and cuff musculature for *at least* 6 to 8 weeks unless advised otherwise.

Stretching Exercises

- Avoid vigorous stretching, the use of contract–relax procedures or grade III joint mobilizations for at least 6 weeks and often for 12 weeks postoperatively to give time for the repaired tendon(s) to heal and become strong.
  - If the supraspinatus or infraspinatus was repaired, initially avoid end-range stretching into internal rotation.
  - If the subscapularis was repaired, initially avoid end-range stretching into external rotation.
  - If the deltoid was detached and repaired, initially avoid end-range shoulder extension, adduction, and horizontal adduction.

Activities of Daily Living

- Wait until about 6 weeks after a mini-open or arthroscopic repair and 12 weeks after a traditional open repair before using the operated arm for light functional activities.
- After repair of a large or massive cuff tear, avoid use of operated arm for functional activities that involve heavy resistance (pushing, pulling, lifting, carrying heavy loads) for 6 to 12 *months* postoperatively.

Exercise: Maximum Protection Phase

The priorities during the initial phase of rehabilitation are protection of the repaired tendon, which is at its weakest approximately 3 weeks after repair,[179] and prevention of the adverse effects of immobilization through early passive motion. In almost all instances, during the first few days after surgery the immobilization (sling or splint) is removed for brief sessions of passive or assisted ROM within limited (protected) and comfortable ranges (see Table 17.4).

The maximum protection phase extends for as little as 3 to 4 weeks after a fully arthroscopic or mini-open repair of small or medium tears or as long as 6 to 8 weeks after repair of large or massive tears. After a fully arthroscopic repair of a small or medium cuff tear, every effort is made to attain nearly full passive shoulder ROM, particularly elevation and external rotation, by 6 to 8 weeks postoperatively.[56,111,190]

Goals and interventions. The following goals and selected interventions combined with the appropriate use of pain medication are initiated during the maximum protection phase.*

- ***Control pain and inflammation.***
  - Periodic use of ice.
  - Arm support for comfort.
  - Cervical spine ROM and shoulder relaxation exercises.
  - Grade I oscillations of the GH joint.
- ***Prevent loss of mobility of peripheral joints.***
  - Assisted ROM of the elbow.
  - Active ROM of the wrist and hand.
- ***Prevent shoulder stiffness/restore shoulder mobility.***
  - Pendulum exercises typically the first postoperative day or when the immobilizer may be removed for exercise.
  - Passive ROM of the shoulder within safe and pain-free ranges. Initially perform exercises in the supine position; begin with elevation of the arm and external rotation in the plane of the scapula.
  - Self-assisted ROM using the opposite hand or a wand for control by 1 to 2 weeks for patients with repairs of small to medium tears and about 2 weeks later for patients with repairs of large tears.
  - Active control of the shoulder with assistance as needed from therapist or family member. With the patient lying in the supine position, place the arm in 90° of shoulder flexion if pain-free. In this position the effect of gravity on the shoulder musculature is minimal. This position has been called the "balance point position" of the shoulder.[50] Help the patient control the shoulder while moving to and from the balance point position, making small arcs and circles with the arm.
  - Active shoulder ROM by the latter part of this phase for small tears and as symptoms permit, initially supine with the elbow flexed, later in a semi-reclining position with the elbow less flexed.

*See references 8, 24, 37, 50, 52, 56, 71, 111, 190, 191.

PRECAUTION: Use only passive and no assisted ROM for 6 to 8 weeks for a repair of a massive cuff tear or after a traditional open repair with deltoid detachment.[37,191]

- ***Prevent or correct postural deviations.***
  - Posture training and exercises to prevent excessive thoracic kyphosis (see Chapters 14 and 16).
- ***Develop control of scapulothoracic stabilizers.***
  - Active movements of the scapula.
  - Submaximal isometrics to isolated scapular muscles.[103] See that the operated arm is supported but not bearing weight to avoid excessive tension in repaired GH musculature.
  - Side-lying scapular protraction/retraction to emphasize control of the serratus anterior.
- ***Prevent inhibition and atrophy of GH musculature.***
  - Low-intensity muscle-setting exercises (against minimal resistance). Setting exercises should not provoke pain in a healing cuff tendon. Begin as early as 1 to 3 weeks postoperatively depending on the size of the tear and quality of the repair.[37,50,52]

PRECAUTION: Recommendations for the safest position of the shoulder in which to begin isometric training of the GH musculature after cuff repair are inconsistent in the literature. One suggestion is to start in a position that creates minimal tension on the repaired cuff tendons (shoulder internally rotated and flexed and abducted to about 45° and elbow flexed).[52] Another option is to place the shoulder in 100° to 110° of flexion and 10° to 20° of horizontal abduction. In this position the deltoid creates a compression force on the head of the humerus into the glenoid fossa, thus diminishing the superior sheer forces generated by the deltoid than when the arm is in less flexion. As the strength of the cuff muscles improves during the later phases of rehabilitation, isometric activities can be performed with the arm positioned in less shoulder flexion.

Exercise: Moderate Protection Phase

The focus of the second phase of rehabilitation is to begin to develop strength, endurance, and neuromuscular control of the shoulder while continuing to attain full or nearly full, pain-free shoulder motion. Emphasis is placed on developing control of the scapular stabilizers and rotator cuff muscles.

For a patient with a repair of a small or medium tear, this phase begins around 4 to 6 weeks postoperatively and extends an additional 6 weeks. For most patients, strengthening exercises typically begin around 8 weeks postoperatively. This phase may begin as late as 12 weeks for a patient with a repair of a large or masssive tear.

Criteria for progression. Criteria to advance to this phase are a well healed incision, minimal pain with assisted shoulder motions, and progressive improvement in ROM. The extent of improvement at just 6 weeks or so after surgery is dependent on factors such as the type and quality of the repair, the size of the tear, and preoperative ROM.